when the concentration of solute is very low, the dilute solution behaves ideally. If the two components present in the solution (A and B) are identical in size, structure, and having almost similar intermolecular attractive forces between them (i.e. between A-A, B-B and B-A) and then the solution tends to behave like an ideal solution.

For an ideal solution

1. There is no change in the volume on mixing the two components (solute & solvents). ($\Delta V_{mixing} = 0$)
2. There is no exchange of heat when the solute is dissolved in solvent ($\Delta H_{mixing} = 0$).
3. Escaping tendency of the solute and the solvent present in it should be same as in pure liquids.

Examples for ideal solutions: (Benzene & Toluene) ; (n-hexane & n-heptane) ; (Ethyl bromide & Ethyl iodide) ; (Chlorobenzene & Bromobenzene).

9.8.2 Non-ideal solutions

The solutions which do not obey Raoult's law over the entire range of concentration, are called non-ideal solutions. For a non-ideal solution, there is a change in the volume and enthalpy upon mixing. i.e. $\Delta H_{mixing} \neq 0$ & $\Delta V_{mixing} \neq 0$. The deviation of the non-ideal solutions from the Raoult's law can either be positive or negative.

Non-ideal solutions - positive deviation from Rauolt's Law:

The nature of the deviation from the Rauolt's law can be explained in terms of the intermolecular interactions between solute (B) and solvent (A). Consider a case in which the intermolecular attractive forces between A and B are weaker than those between the molecules of A (A-A) and molecules of B (B-B). The molecules present in such a solution have a greater tendency to escape from the solution when compared to the ideal solution formed by A and B, in which the intermolecular attractive forces (A-A, B-B, A-B) are almost similar. Consequently, the vapour pressure of such non-ideal solution increases and it is greater than the sum of the vapour pressure of A and B as predicted by the Raoult's law. This type of deviation is called positive deviation.

Here, $p_A > p^\circ_A x_A$ and $p_B > p^\circ_B x_B$.

Hence $p_{total} > p^\circ_A x_A + p^\circ_B x_B$ ----------(9.19)

Let us understand the positive deviation by considering a solution of ethyl alcohol and water. In this solution the hydrogen bonding interaction between ethanol and water is weaker than those hydrogen bonding interactions amongst themselves (ethyl alcohol-ethyl alcohol and water-water interactions). This results in the increased evaporation of both components (H_2O and C_2H_5OH) from the aqueous solution of ethanol. Consequently, the vapour pressure of the solution is greater than the vapour pressure predicted by Raoult's law. Here, the mixing process is endothermic i.e. $\Delta H_{mixing} > 0$ and there will be a slight increase in volume ($\Delta V_{mixing} > 0$).

Examples for non-ideal solutions showing positive deviations: Ethyl alcohol & cyclohexane, Benzene & acetone, Carbon tetrachloride & chloroform, Acetone & ethyl alcohol, Ethyl alcohol & water.

Unit 9.indd 46 23-12-2021 17:00:56